Phase 2: Information Design & IA Mapping

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Empirical Card Sorting Experiment

Card Deck Preparations

I created 18 physical index cards containing core content items for weber state university canvas. The list included: Dashboard, Courses, Inbox, Calendar, History, Files, Groups, People, Grades, Notebook, Media, Pages, Syllabus, Announcements, Discussions, Quizzes, Modules, and Assignments.

Participant Execution

I invited 3 users to organize the cards into groups that felt intuitive, asked them to label their groups and then asked them to explain why they organized things how they did. The first person I asked was a student within the child education department, he organized them by how often he used each component. The second was an arts major based on what they reminded her off. The third was a teacher who organized it based on who uses it.

Data Synthesis & Analysis

The first set of groupings is done by how often he uses them.

Most Used	Semi Useful	Why do we have these
Courses	Syllabus	Pages
Modules	Grades	Discussions
Calendar	Dashboard	Announcements
	Media	Files
	Assignments	Notebook
	Quizzes	People
		Inbox
		Groups
		History

The second set of groupings was based on what they reminded her of.

School	Computers	Social	Notebook
Assignments	Calendar	Groups	Notebook
Grades	Files	People	Pages
Syllabus	Dashboard	Media	Discussions
Courses	Inbox		Announcements
Quizzes			
Modules			
History			

The Third set of grouping were done by who was using them and her opinions of how they are used.

Functions of Canvas	Teacher Used	Not Good Enough	Student Used
Courses	Groups	Discussions	Media
Dashboard	People	Announcements	Notebook
Inbox	Grades	Syllabus	Files
	Assignments	Pages	History
	Modules		Calendar
	Quizzes		

Sorting by how they use the pages was very common, which lead to things like assignments, quizzes, and modules being grouped together frequently. Interestingly though discussion is functionally similar to quizzes and Assignments as a way to do work on canvas they were never grouped together as discussions are more often paired with Pages and Groups which makes sense as discussions require communicating with your peers or are an easy way to communicate within a group. Another common grouping is based on components that are not commonly used, such as Discussions, Announcements, Pages. This group stands out as some of the things on the pages come up often but very rarely will users go to that page itself. People and Groups shows how people connect the two in function.

One of the points of disagreement comes from how a teacher vs a student sees groups, a teacher views it as important tool while a student may never use it. The second user stands apart from the rest due to focusing on what they remind her of rather than how they are used.

Revised Sitemap & Information Architecture Blueprint.

Primary navigation is the primary Global Navbar.

Global Side Bar: Dashboard/Courses, Inbox, Calendar, Files, History

Dashboard has a button on each course card that leads to grades, and announcements for quicker navigation.
Calendar has links to each assignment as they show up on the page.
History records pages you were on and as links back to them as well as shows recent actions such as submitting on time or late.

Top Bar: Contains bread crumb to allow for easy returning to higher pages.
"Dashboard"

Secondary navigation is the course side bar.

Course Side Bar: Modules, Announcements, Grades, Groups, Syllabus

Grades have links to Assignments, Quizzes, and Discussions to allow for direct review.

Syllabus has in line links to modules, assignments, quizzes, discussions, and/or grades if the teacher sets them up to review active upcoming expectations.

Top Bar: Contains bread crumb to allow for easy returning to higher pages.
"Dashboard > Class"

Utility Navigation – Subpage types with defined format but fall under a different category.

Files: Media, Pages, Notebook.

Modules: Assignments, Quizzes, Discussions.

Has links to files based on teacher what connections the teacher provides to given course files or outside Canvas Page Recourses.

Groups: People, Discussions.

Top Bar: Contains bread crumb to allow for easy returning to higher pages.
"Dashboard > Class > Modules > Assignment Name"

Low-Fidelity Wireframes & User Flows

Core Task Selection

I chose to audit the Canvas Home Page, and locating a File.

The Home Page is too reliant on a skilled teacher to set up for a student and can cause dissonance or confusion in the student. I chose to revise the home page by allowing a student to customize their theirs. When they first access the course, it will be set up as the default chosen by the teacher, where in which a student can then make changes by adding or removing widgets from the page. There are many premade widgets for each part of canvas that a student might want to be able to review all in one place, such as modules, upcoming assignments, grades, announcements, View course component link buttons, Todo, groups, and even custom teacher made widgets that contain information, buttons, links or images.

Files, Media, Pages, and Notebook all serve similar functions of contains information saved by a teacher or student in one place, however they are all scattered across various navigation systems. My revision is to group all under a single page on the global navigation bar, “Files” and have it been accessible by the Global Navbar. Within the File Page there are sections each holding the functions of each of the above files as based on the user associations found during the card sorting experiment. There is also a component on the right side that allows you to show or hide the various components.

At the top there is the Files explorer which shows a list of files with “My Files” at the top, if you chose this you can review all files you have uploaded or used in canvas. There is a hard coded locked folder Called “Submissions” that has subfolders for all classes that you have submitted files in.

The next is Media, which is any video you have uploaded and tagged. These can be shared or accessed externally using a link.

Notebook allows a student to highlight content on any class page and save it in one location. It is sorted by class then page and can be filtered and pinned based on how the student tags the information.

Pages are a list of all pages the teacher has saved for the students to access.

User Flow Diagramming

Home Page Customization.

Sign In to Canvas

- Land on the dashboard.
- Choose course from the dashboard
- Land on the Home Page
- User reviews current home page and clicks on the “Customize Home Page” button on the top right of the page.
- Page changes to show customize features, such as where the “Customize Home Page” button was there is now two buttons “Cancel”
  and “Save Changes” buttons. there is also the “+ add widget” button above, between, and bellow each currently present widget in two columns. The first column is the biggest containing the main content of the page in the middle. The second is the same size as both navbars on the left other side and will have the widgets display in their thinner form.
    - Click on the “+ Add Widget” button
        - Opens the widget menu that has all currently existing widgets the teacher has given you access to. This includes any custom ones made by them, if they know how, that they have given you permission for. Each widget in the list has the name of the widget, a description of what it is, if it was made by the teacher, and a button that either says “Add” which you press to choose the widget you wish to add or “Added” which means that widget already is on the home page.
        - Click Add Widget
        - Widget is added to the home page below the button pressed, their will now be a new “+ Add Wiget” button below it so their remains exactly one of the buttons above, between, and below each component.
    - Clicks “Cancel” Button.
        - If any changes have previously been making pops an error message asking “are you sure you wish to leave without saving, changes will be lost” with two buttons “Yes” or “No”.
            - Press “Yes”: Exit Customization mode back to home screen without saving changes.
            - Press “No”: Return to Customization mode.

        - Exits the home page customization mode without saving any changes
    - Clicks on the two columns of three dots on the left to the trashcan in the top right of the widget.
        - Pulls up movement options to move the existing widget on the page.
            - Options are Move to the Top, Move Up, Move down, Move to bottom, Move left top, Move left bottom, Move right top, Move right bottom. Each one moves to the selected Wiget to the described location. For example, Move Up moves the widget one position up. If the widget is already in the position the relative buttons will be locked.
    - Clicks “Save Changes” Button
        - Saves new layout and leaves customization mode.
    - Click on the trashcan on top right of a widget.
        - Removes the widget from the layout.

Locate Class File

Sign In to Canvas

- Land on the dashboard.
- Press “File” Button on global navbar.
- Opens File Explorer Page
- Check what options are being shown on the page
    - If File is unchecked, press the square to check it.
- Locate course name in files section. Select it.
    - This should open the courses files, all the files made available by the teacher of the course.
    - Navigate internal folders as needed to locate the chosen File.
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Wireframe Deck Creation

See *Hand Drawn Prototypes – Files and Home.pdf* to review Prototypes.

Accessibility First & Semantic Navigation Planning

Semantic HTML Structuring

<Header Class= “Top-Bar”> - Primary contains bread crumb showing where you are in the site. It also contains collapse button that hides <Nav class= “Course-Nav”> and any additional page manipulation buttons, such as <button id= “Customize-Home”>. When in Customize mode, it also contains the save and cancel buttons.

<Nav class= “Course-Nav”> - Contains all <Buttons> for secondary navigation within a course.

> Modules, Announcements, Grades, Groups, Syllabus, plus any additional class pages as set up by the instructors.

<Main> - Contains all sections. It also contains customization components such as “+ Add Widget” buttons

> <Section id= “Module-widget” class= “Home-Widget”> - Contains the Module Widget showing the modules from the moule page.
> <Section id= “Grade-widget” class= “Home-Widget”> - contains the Grade widget showing the grade of the user in the course and its breakdown.
>
> <Section id= “Custom-widget []”> - This contains a custom widget made by a teacher for their course.
>
> … etc. – Any widgets I have not thought of at this moment that is made by the teacher or by others.
>
> <Display id= “Course Widget”> - Contains popup list off all current available widgets a student may use to customize their page.

File Page

<Nav Class= “Global-Nav”> - Contains all <buttons> that are in the primary navigation.

> Dashboard, Inbox, Calendar, Files, and History.

<Header Class= “Top-Bar”> - Primary contains bread crumb showing where you are in the site. It also contains collapse button that hides <Nav class= “Course-Nav”>

<Nav class= “Course-Nav”> - Contains all <Buttons> for secondary navigation within a course.

Modules, Announcements, Grades, Groups, Syllabus, plus any additional class pages as set up by the instructors.

<aside id= “Visibility-Toggle” class= “Visibility-Toggle>

<Main> - Contains all sections, as well as the visibility toggle <Div>.

<Section id= “File-Explorer”> - contains the file explorer functionality, allows the user to move through user and teacher provided files. Allows a user to upload and organize their own files on canvas and download teacher provided ones. Can be hidden by the <aside id= “Visibility-Toggle” class= “Visibility-Toggle>.

<Section id= “Media”> - contains the media functionality. Allows for the user to tag, manipulate and share videos and images that are available to use in discussions or textboxes. <aside id= “Visibility-Toggle” class= “Visibility-Toggle>.

<Section id= “Notebook”> - contains the Notebook functionality allowing the user to view and mange highlighted content on other pages in courses. Can be hidden by the <aside id= “Visibility-Toggle” class= “Visibility-Toggle>.

<Section id= “Pages”> - contains links to all outside pages the teacher has saved for student use. Can be hidden by the <aside id= “Visibility-Toggle” class= “Visibility-Toggle>.

Keyboard & ARIA Blueprint

Course Home Page

Starts on the Global Navbar, tab moves down the list, it then moves across the bread crumb in the top bar, highlighting the customize button. It Moves on to the course navbar before moving into main. Move through each widget in the middle before moving to the ones on the left

Each component above is labeled appropriately using aria-label=””. The navbars both have aria-current= “Page” on the appropriate button, as well as aria-role= “navigation” for each <nav>. The Collapse Course Navbar button as the additional aria-property of aria-expanded= “True”. The main page also has aria-role= “main”. During Customization the move section has aria-live= “polite” as does the widget choice dialog box. Also, each widget is labeled on its title to make sure it is clear to the screen reader what each one is. Whatever is on the top left, top right, bottom left, and bottom right also have additional labels in the screen reader explaining its position. After it moves it re-reads its new position in the column it is in. For example, it can have the aria-label= “Modules, Position 1 on the

right” or “Grades, position 2 on the left” etc.. This would have to be programed via JavaScript, it is done at the same time as the element is moved.

File Page

Starts on the Global Navbar, tab moves down the list, it then moves across the bread crumb in the top bar, then moves onto Main. Move to the visibility controls before moving through each section that is visible in the middle.

Each component above is labeled appropriately using aria-label=””. The navbars both have aria-current= “Page” on the appropriate button, as well as aria-role= “navigation” for each <nav>. Each visibility button has the aria-property of aria-expanded= “true”. The main page also has aria-role= “main”.

Post-Mortem Reflection (Metacognition)

1. How did the empirical data from your card sorting experiment challenge your initial assumptions about how the system's navigation should be structured?

   I figured the navigation was already pretty good, but doing this I realized that there were several pages that felt very similar to one another. Hence why I chose to upgrade the home page to make it feel more distinct to the student. This is also why I thought to group all the information pages together under the new file tab on the Global Navbar, so these similar pages each could be kept together so users didn't need to hunt for them.

2. Describe how your annotated ARIA landmarks and keyboard tab sequences ensure that screen reader users navigate their revised information architecture as efficiently as sighted users.

   As I made paper prototypes, I made sure to keep track as I made them what would need what annotation on the pages with ARIA. Then I wrote out the above to keep track of what I would need to put for each major component. For example, if it is a button that hides something I noted down needing to add aria expanded. On navigation bars I wrote about using aria-current= "page" and aria-role= "navigation". I want my redesigned prototypes to be quickly updated with appropriate landmarks. As I set up how things would move through the page, I made sure to organize the page from top to bottom with the order I would want to move through the page. For example I set up the file page to have the visibility toggles appear before any other content despite it being next to the complete content, this way a person navigating would have the ability to control what pages they wish to see without having to loop through the whole page to see what changed. I took the time to keep them in mind as I worked so that hey can use the system just as easily as I can. I didn't want any confusion about what something does or why it exists.